



SIMATS
ENGINEERING



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Savitribha Institute of Medical and Technical Sciences
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CLOUD NETWORKING THROUGH THE THE OSI ARCHITECTURE

CSA0704-COMPUTER NETWORKS

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INTRODUCTION

Cloud Networking

- ❑ Connects users, devices and cloud services through the internet
- ❑ Supports communication between clients and cloud data centers
- ❑ Enables on-demand access to storage, applications and computing resources

OSI Architecture

- ❑ A 7-layer reference model for network communication
- ❑ Standardizes how data is transmitted across networks

OSI LAYERS IN CLOUD NETWORKING

Application

- ❑ Provides applications and services to users

Presentation

- ❑ Encrypts, decrypts and compresses data

Session

- ❑ Establishes and maintains communication session

Transport

- ❑ Provides reliable data delivery using TCP/UDP

Data link

- ❑ Transfers data frames between network devices

Physical

- ❑ Transmits bits through cables, fiber, optics or wireless media

DATA FLOW IN CLOUD NETWORKING

Step-by-Step Process

1. User sends a request to a cloud application.
2. Application layer processes the request.
3. Presentation layer encrypts the data.
4. Session layer establishes communication.
5. Transport layer divides data into segments.
6. Network layer routes packets to the cloud server.
7. Data Link and Physical layers transmit data.
8. Cloud server processes the request and returns the response

APPLICATIONS OF CLOUD NETWORKING

- ❑ **Cloud Storage:** Store, access, and share files online (Google Drive, OneDrive, Dropbox).
- ❑ **Web Hosting:** Host websites and web applications on cloud servers.
- ❑ **Video Streaming:** Deliver high-quality video content through platforms like YouTube and Netflix.
- ❑ **Online Collaboration:** Enable teams to work together using tools like Google Workspace and Microsoft 365.
- ❑ **Virtual Machines:** Create and manage virtual servers without physical hardware.
- ❑ **Database Hosting:** Store and manage large amounts of data securely in the cloud.
- ❑ **Disaster Recovery:** Backup and restore data quickly during system failures.



THANK YOU